International Institute of Information Technology, Hyderabad
MA4.101-Real Analysis (Monsoon-2026)

Practice Problems with Solutions

Chapters 7 and 9: Series, and subsets of the real line

Definitions and facts assumed

The solutions below are self-contained: every result used in a proof is stated where it is used. The following definitions and facts are assumed throughout and are not reproved.

Definitions.

(D1) A sequence $(a_n)_{n=m}^{\infty}$ of reals *converges to* a real L iff for every real $\varepsilon > 0$ there is an $N \geq m$ such that $|a_n - L| \leq \varepsilon$ for all $n \geq N$; we then write $L = \lim_{n\to\infty} a_n$. A sequence converging to no real number is said to *diverge*.

(D2) Given reals $a_m, a_{m+1}, \ldots$, the *formal infinite series* $\sum_{n=m}^{\infty} a_n$ has *partial sums* $S_N := \sum_{n=m}^{N} a_n$ for $N \geq m$. If $(S_N)_{N=m}^{\infty}$ converges to a real L, the series is *convergent* with *sum* L, and we write $L = \sum_{n=m}^{\infty} a_n$; otherwise the series is *divergent* and no real value is assigned to it.

(D3) $\sum_{n=m}^{\infty} a_n$ is *absolutely convergent* iff the series $\sum_{n=m}^{\infty} |a_n|$ is convergent.

(D4) *(Limit superior and inferior.)* Suprema and infima here are taken in the extended reals $\mathbb{R}^* := \mathbb{R} \cup \{+\infty, -\infty\}$, ordered by $-\infty < x < +\infty$ for every real x: for $E \subseteq \mathbb{R}$ non-empty and bounded above, $\sup(E)$ is its least upper bound; $\sup(E) := +\infty$ if E is non-empty and not bounded above; $\sup(\emptyset) := -\infty$; and $\inf(E) := -\sup(-E)$, where $-E := \{-x : x \in E\}$. For a sequence $(a_n)_{n=m}^{\infty}$ and $N \geq m$ put $a_N^+ := \sup\{a_n : n \geq N\}$ and $a_N^- := \inf\{a_n : n \geq N\}$; then

$$\limsup_{n\to\infty} a_n := \inf\{\, a_N^+ : N \geq m \,\}, \qquad \liminf_{n\to\infty} a_n := \sup\{\, a_N^- : N \geq m \,\}.$$

(D5) Let $X \subseteq \mathbb{R}$, let $\varepsilon > 0$ and let $x \in \mathbb{R}$. Then x is *ε-adherent* to X iff there is $y \in X$ with $|x - y| \leq \varepsilon$; and x is an *adherent point* of X iff it is ε-adherent

to X for every $\varepsilon > 0$. The *closure* $\overline{X}$ is the set of all adherent points of X.

(D6) $E \subseteq \mathbb{R}$ is *closed* iff $E = \overline{E}$, that is, iff E contains all of its adherent points.

(D7) $X \subseteq \mathbb{R}$ is a *bounded set* iff $X \subseteq [-M, M]$ for some real $M > 0$.

Facts.

(F1) $|a + b| \leq |a| + |b|$, $|ab| = |a|\,|b|$, $|-a| = |a|$, and $|a| = 0$ if and only if $a = 0$.

(F2) *(Archimedean property.)* For every real x there is a positive integer N with $x \leq N$.

(F3) *(Limit laws.)* If $a_n \to L$ and $b_n \to M$ then $a_n + b_n \to L + M$, $a_n b_n \to LM$, and $c\,a_n \to cL$ for every real c; and if moreover $M \neq 0$ and every $b_n \neq 0$, then $a_n / b_n \to L/M$.

(F4) Limits preserve non-strict inequalities: if $a_n \to L$ and $a_n \geq c$ for all n, then $L \geq c$; and likewise with $\leq$.

(F5) Every convergent sequence of reals is bounded.

(F6) *(Suprema and infima.)* For $E \subseteq \mathbb{R}^*$: every $x \in E$ satisfies $\inf(E) \leq x \leq \sup(E)$; if M is an upper bound for E then $\sup(E) \leq M$; and if M is a lower bound for E then $\inf(E) \geq M$. If $E \subseteq \mathbb{R}$ is non-empty and bounded above then $\sup(E)$ is a real number and for every $\varepsilon > 0$ there is $y \in E$ with $y > \sup(E) - \varepsilon$; symmetrically, if E is non-empty and bounded below then $\inf(E)$ is real and for every $\varepsilon > 0$ there is $y \in E$ with $y < \inf(E) + \varepsilon$.

(F7) *(Properties of* $\limsup$ *and* $\liminf$*.)* For real sequences $(a_n)_{n=m}^{\infty}$ and $(b_n)_{n=m}^{\infty}$:

- (i) $\liminf_{n\to\infty} a_n \leq \limsup_{n\to\infty} a_n$;
- (ii) if $x > \limsup_{n\to\infty} a_n$ there is $N \geq m$ with $a_n < x$ for all $n \geq N$; and if $y < \liminf_{n\to\infty} a_n$ there is $N \geq m$ with $a_n > y$ for all $n \geq N$;
- (iii) for a real c, (a_n) converges to c if and only if $\limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n = c$;
- (iv) if $a_n \leq b_n$ for all $n \geq m$ then $\limsup_{n\to\infty} a_n \leq \limsup_{n\to\infty} b_n$ and $\liminf_{n\to\infty} a_n \leq \liminf_{n\to\infty} b_n$.

(F8) $\lim_{n\to\infty} c^{1/n} = 1$ for every real $c > 0$.

(F9) For a real $x \geq 0$ and an integer $n \geq 1$, the number $x^{1/n}$ is the unique non-negative real whose nth power is x. Moreover $x > y \geq 0$ if and only if $x^{1/n} > y^{1/n}$, and $(xy)^{1/n} = x^{1/n}y^{1/n}$.

(F10) For every real x there is a unique integer $\lfloor x \rfloor$ with $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$.

(F11) The partial sums of $\sum_{k=1}^{\infty} 1/k$ are not bounded above.

(F12) *(Alternating series test.)* Let $(a_n)_{n=m}^{\infty}$ be non-negative and decreasing, thus $a_n \geq 0$ and $a_n \geq a_{n+1}$ for every $n \geq m$. Then $\sum_{n=m}^{\infty}(-1)^n a_n$ converges if and only if $a_n \to 0$.

Part A consists of the results proved in lectures, set as exercises. Part B contains further questions built on them.

Part A1. Series

Question (1). Let $\sum_{n=0}^{\infty} a_n$ be an absolutely convergent series of real numbers such that $\sum_{n=0}^{\infty} |a_n| = 0$. Show that $a_n = 0$ for every natural number n.

By (D3) and (D2), the partial sums $S_K := \sum_{n=0}^{K} |a_n|$ form a sequence converging to 0.

Step 1: (S_K) is non-negative and increasing. Each $|a_n| \geq 0$, so $S_K \geq 0$, and $S_{K+1} - S_K = |a_{K+1}| \geq 0$.

Step 2: $S_K \leq 0$ for every K. Fix K. By Step 1, $S_j \geq S_K$ for every $j \geq K$. Discarding the first K terms changes neither convergence nor limit, so (F4) applied to $(S_j)_{j \geq K}$ with the constant S_K gives

$$0 \;=\; \lim_{j \to \infty} S_j \;\geq\; S_K.$$

Step 3: conclusion. By Steps 1 and 2, $S_K = 0$ for every natural K. Hence $|a_0| = S_0 = 0$, and for $n \geq 1$,

$$|a_n| \;=\; S_n - S_{n-1} \;=\; 0 - 0 \;=\; 0.$$

By (F1), $|a_n| = 0$ gives $a_n = 0$, for every natural number n.

Remark. The conclusion fails without the absolute values: $\sum_n a_n = 0$ certainly does not force every a_n to vanish, as $a_0 = 1$, $a_1 = -1$ and $a_n = 0$ otherwise shows. Non-negativity of the terms $|a_n|$ is the only property used, and it is used in Step 1.

Question (2). Let $(c_n)_{n=m}^{\infty}$ be a sequence of positive numbers. Show that

$$\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \;\le\; \liminf_{n\to\infty} c_n^{1/n} \;\le\; \limsup_{n\to\infty} c_n^{1/n} \;\le\; \limsup_{n\to\infty} \frac{c_{n+1}}{c_n}.$$

(Here $m \ge 1$, so that $c_n^{1/n}$ is defined.)

Write $r_n := c_{n+1}/c_n$, which is positive, and $t_n := c_n^{1/n}$, which is positive by (F9) since $c_n > 0$. Put

$$\ell := \liminf_{n\to\infty} r_n, \qquad L := \limsup_{n\to\infty} r_n.$$

The middle inequality is (F7)(i) applied to (t_n), so two inequalities remain.

A preliminary observation. Both ℓ and L are ≥ 0. Indeed 0 is a lower bound for $\{r_n : n \ge N\}$ for every N, so the quantity r_N^- of (D4) satisfies $r_N^- \ge 0$ by (F6); hence $\ell = \sup\{r_N^-\} \ge 0$ by (F6) again, and $L \ge \ell \ge 0$ by (F7)(i). The same argument gives $\liminf_{n\to\infty} t_n \ge 0$ and $\limsup_{n\to\infty} t_n \ge 0$.

The last inequality: $\limsup_{n\to\infty} t_n \le L$. If $L = +\infty$ there is nothing to prove, since every extended real is $\le +\infty$. So assume L is a real number, necessarily $L \ge 0$.

Let $\varepsilon > 0$. Since $L + \varepsilon > L$, (F7)(ii) supplies an $N \ge m$ with

$$r_n \;<\; L + \varepsilon \qquad \text{for all } n \ge N,$$

that is, $c_{n+1} \le c_n(L+\varepsilon)$ for all $n \ge N$, using $c_n > 0$. By induction on n,

$$c_n \;\le\; c_N(L+\varepsilon)^{n-N} \qquad \text{for all } n \ge N:$$

this holds at $n = N$, and if it holds at n then $c_{n+1} \le c_n(L+\varepsilon) \le c_N(L+\varepsilon)^{n+1-N}$.

Put $A := c_N(L+\varepsilon)^{-N}$, a positive real since $L + \varepsilon > 0$ and $c_N > 0$. The display reads $c_n \le A(L+\varepsilon)^n$ for $n \ge N$. Taking nth roots and using (F9),

$$t_n \;=\; c_n^{1/n} \;\le\; \big(A(L+\varepsilon)^n\big)^{1/n} \;=\; A^{1/n}\big((L+\varepsilon)^n\big)^{1/n} \;=\; A^{1/n}(L+\varepsilon)$$

for all $n \ge N$; the first step uses the monotonicity in (F9) in its non-strict form, and the last uses that $L + \varepsilon$ is a non-negative real whose nth power is $(L+\varepsilon)^n$.

By (F8), $A^{1/n} \to 1$, so by (F3) the sequence $A^{1/n}(L+\varepsilon)$ converges to $L+\varepsilon$, and therefore has limit superior $L+\varepsilon$ by (F7)(iii). Comparing with (F7)(iv),

$$\limsup_{n\to\infty} t_n \;\le\; \limsup_{n\to\infty} A^{1/n}(L+\varepsilon) \;=\; L+\varepsilon.$$

This holds for every $\varepsilon > 0$. Taking $\varepsilon := 1$ shows $\limsup_{n\to\infty} t_n \le L + 1 < +\infty$, and we noted $\limsup_{n\to\infty} t_n \ge 0$, so $\limsup_{n\to\infty} t_n$ is a real number. If

$\limsup_{n\to\infty} t_n > L$, then $\varepsilon := (\limsup_{n\to\infty} t_n - L)/2 > 0$ would give

$$\limsup_{n\to\infty} t_n \;\le\; L + \frac{\limsup_{n\to\infty} t_n - L}{2} \;=\; \frac{L + \limsup_{n\to\infty} t_n}{2} \;<\; \limsup_{n\to\infty} t_n,$$

which is absurd. Hence $\limsup_{n\to\infty} t_n \le L$.

The first inequality: $\ell \le \liminf_{n\to\infty} t_n$. If $\ell = 0$ this is the observation above, that $\liminf_{n\to\infty} t_n \ge 0$. So assume $\ell > 0$, and let y be any real with $0 < y < \ell$.

By (F7)(ii), there is $N \ge m$ with $r_n > y$ for all $n \ge N$, that is, $c_{n+1} \ge c_n y$. By induction exactly as before,

$$c_n \;\ge\; c_N y^{n-N} \;=\; B y^n \qquad \text{for all } n \ge N, \qquad B := c_N y^{-N} > 0.$$

Taking nth roots with (F9) gives $t_n \ge B^{1/n} y$ for $n \ge N$. By (F8) and (F3), $B^{1/n} y \to y$, so its limit inferior is y by (F7)(iii), and (F7)(iv) gives

$$\liminf_{n\to\infty} t_n \;\ge\; \liminf_{n\to\infty} B^{1/n} y \;=\; y.$$

So $\liminf_{n\to\infty} t_n \ge y$ for every real y with $0 < y < \ell$.

If $\ell = +\infty$, this says $\liminf_{n\to\infty} t_n \ge y$ for every positive real y, which forces $\liminf_{n\to\infty} t_n = +\infty = \ell$. If ℓ is a positive real and $\liminf_{n\to\infty} t_n < \ell$, then $\liminf_{n\to\infty} t_n \ge y$ fails for $y := \max(\liminf_{n\to\infty} t_n, 0) + \frac{1}{2}\big(\ell - \max(\liminf_{n\to\infty} t_n, 0)\big)$, a real number strictly between 0 and ℓ and strictly greater than $\liminf_{n\to\infty} t_n$ — a contradiction. Hence $\ell \le \liminf_{n\to\infty} t_n$.

Question (3). Show that $\lim_{n\to\infty} n^{1/n} = 1$.

We use the following chain of inequalities, valid for every sequence $(c_n)_{n=1}^{\infty}$ of positive numbers:

$$\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \;\le\; \liminf_{n\to\infty} c_n^{1/n} \;\le\; \limsup_{n\to\infty} c_n^{1/n} \;\le\; \limsup_{n\to\infty} \frac{c_{n+1}}{c_n}. \qquad (\dagger)$$

Apply it to $c_n := n$ for $n \ge 1$, a sequence of positive numbers.

The ratios converge to 1. We have $c_{n+1}/c_n = (n+1)/n = 1 + 1/n$. Now $1/n \to 0$: given $\varepsilon > 0$, (F2) supplies a positive integer N with $1/\varepsilon \le N$, and then $|1/n - 0| = 1/n \le 1/N \le \varepsilon$ for all $n \ge N$. So by (F3), $c_{n+1}/c_n \to 1$.

Hence both ratio quantities equal 1. By (F7)(iii),

$$\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \;=\; \limsup_{n\to\infty} \frac{c_{n+1}}{c_n} \;=\; 1.$$

Conclusion. With these values, (†) reads

$$1 \;\leq\; \liminf_{n\to\infty} n^{1/n} \;\leq\; \limsup_{n\to\infty} n^{1/n} \;\leq\; 1,$$

so $\liminf_{n\to\infty} n^{1/n} = \limsup_{n\to\infty} n^{1/n} = 1$, and (F7)(iii) in the other direction gives $\lim_{n\to\infty} n^{1/n} = 1$.

Part A2. Subsets of the real line

Question (4). Let I be a non-empty set (possibly infinite), and for each $\alpha \in I$ let X_α be a closed subset of $\mathbb{R}$. Show that the intersection $\bigcap_{\alpha\in I} X_\alpha$ is also closed.

Since I is non-empty, the intersection is defined. Write $X := \bigcap_{\alpha\in I} X_\alpha$. By (D6) we must show that every adherent point of X lies in X.

So let x be adherent to X, and fix $\alpha \in I$. Let $\varepsilon > 0$. By (D5) there is $y \in X$ with $|x - y| \leq \varepsilon$; and $y \in X$ gives $y \in X_\alpha$. So x is ε-adherent to X_α. As $\varepsilon > 0$ was arbitrary, x is an adherent point of X_α, and since X_α is closed, $x \in X_\alpha$.

As $\alpha \in I$ was arbitrary, $x \in \bigcap_{\alpha\in I} X_\alpha = X$. Hence $\overline{X} \subseteq X$. The reverse inclusion holds for any set, since each $x \in X$ is ε-adherent to X for every ε (take $y := x$). So $X = \overline{X}$ and X is closed.

Question (5). Show that if X is a bounded subset of $\mathbb{R}$, then the closure $\overline{X}$ is also bounded.

By (D7) there is a real $M > 0$ with $X \subseteq [-M, M]$, that is, $|x| \leq M$ for every $x \in X$. We show $\overline{X} \subseteq [-M, M]$, which gives the claim with the same M.

Let $y \in \overline{X}$ and let $\varepsilon > 0$. By (D5) there is $x \in X$ with $|y - x| \leq \varepsilon$, so by (F1),

$$|y| \;=\; |(y - x) + x| \;\leq\; |y - x| + |x| \;\leq\; \varepsilon + M.$$

This holds for every $\varepsilon > 0$. If $|y| > M$, then taking $\varepsilon := (|y| - M)/2 > 0$ gives

$$|y| \;\leq\; M + \frac{|y| - M}{2} \;=\; \frac{|y| + M}{2} \;<\; |y|,$$

which is absurd. Hence $|y| \leq M$, as required.

Remark. The proof shows more than was asked: *every* bound for X is a bound for $\overline{X}$. So passing to the closure never enlarges the smallest available bound.

Question (6). Show that the closure of $\mathbb{N}$ is $\mathbb{N}$, the closure of $\mathbb{Z}$ is $\mathbb{Z}$, the closure of $\mathbb{Q}$ is $\mathbb{R}$, the closure of $\mathbb{R}$ is $\mathbb{R}$, and the closure of the empty set $\emptyset$ is $\emptyset$.

The method. Each assertion is an equality of two sets, so there are two inclusions to check, and they are of very different difficulty.

One inclusion is free. For every $Y \subseteq \mathbb{R}$ we have $Y \subseteq \overline{Y}$: if $x \in Y$ and $\varepsilon > 0$, then the point $y := x$ lies in Y and satisfies $|x - y| = 0 \leq \varepsilon$, so x is ε-adherent to Y; as $\varepsilon > 0$ was arbitrary, x is an adherent point of Y, that is, $x \in \overline{Y}$ by (D5).

The other inclusion is the work. To prove $\overline{Y} \subseteq Y$ we argue by contraposition: we take an arbitrary $x \in \mathbb{R} \setminus Y$ and show it is not an adherent point of Y. Negating (D5), this means producing a *single* witness $\varepsilon > 0$ for which

$$|x - y| \;>\; \varepsilon \qquad \text{for every } y \in Y. \tag{$*$}$$

In each case below the witness is obtained the same way: we measure how far x is from the nearest points of Y, find that this distance is some $d > 0$, and then take $\varepsilon := d/2$, which no point of Y can reach.

For $\mathbb{Q}$ and $\mathbb{R}$ the situation is reversed — there it is $\overline{Y} \subseteq \mathbb{R}$ that is free, and we must instead show every real *is* adherent, which means succeeding for *every* ε rather than failing for one.

$\overline{\mathbb{Z}} = \mathbb{Z}$. Let $x \in \mathbb{R} \setminus \mathbb{Z}$.

Step 1: locate x between consecutive integers. By (F10) there is a unique integer $k := \lfloor x \rfloor$ with

$$k \;\leq\; x \;<\; k + 1.$$

Since $k \in \mathbb{Z}$ and $x \notin \mathbb{Z}$ we have $x \neq k$, so the left inequality is strict:

$$k \;<\; x \;<\; k + 1.$$

Step 2: measure the distance to the nearest integer. Put

$$d \;:=\; \min\big(x - k,\; (k+1) - x\big).$$

Both of the two quantities are strictly positive by Step 1, so $d > 0$.

We claim that $|x - j| \geq d$ for *every* integer j. Since there is no integer strictly between k and $k+1$, every $j \in \mathbb{Z}$ satisfies $j \leq k$ or $j \geq k+1$, and we treat these in turn.

- If $j \le k$, then $j \le k < x$, so $x - j > 0$ and $|x - j| = x - j \ge x - k \ge d$.
- If $j \ge k{+}1$, then $j \ge k{+}1 > x$, so $j{-}x > 0$ and $|x{-}j| = j{-}x \ge (k{+}1){-}x \ge d$.

Step 3: the witness. Take $\varepsilon := d/2 > 0$. By Step 2, $|x - j| \ge d > d/2 = \varepsilon$ for every $j \in \mathbb{Z}$, which is exactly $(*)$. So x is not an adherent point of $\mathbb{Z}$.

As $x \in \mathbb{R} \setminus \mathbb{Z}$ was arbitrary, $\overline{\mathbb{Z}} \subseteq \mathbb{Z}$; with the free inclusion, $\overline{\mathbb{Z}} = \mathbb{Z}$.

$\overline{\mathbb{N}} = \mathbb{N}$. Let $x \in \mathbb{R} \setminus \mathbb{N}$. There are two cases.

Case $x < 0$. Every $n \in \mathbb{N}$ satisfies $n \ge 0 > x$, so

$$|x - n| \;=\; n - x \;\ge\; 0 - x \;=\; -x \;>\; 0.$$

Taking $\varepsilon := -x/2 > 0$ gives $|x - n| \ge -x > \varepsilon$ for every $n \in \mathbb{N}$, which is $(*)$.

Case $x \ge 0$. We first check $x \notin \mathbb{Z}$: if x were an integer, then being also non-negative it would be a natural number, contradicting $x \notin \mathbb{N}$. So Steps 1–2 of the previous part apply to x and produce a $d > 0$ with $|x - j| \ge d$ for every *integer* j. Every natural number is an integer, so $|x - n| \ge d > d/2$ for every $n \in \mathbb{N}$, and $\varepsilon := d/2$ is a witness.

In both cases x is not an adherent point of $\mathbb{N}$, so $\overline{\mathbb{N}} \subseteq \mathbb{N}$ and $\overline{\mathbb{N}} = \mathbb{N}$.

$\overline{\mathbb{Q}} = \mathbb{R}$. By (D5) adherent points are real numbers, so $\overline{\mathbb{Q}} \subseteq \mathbb{R}$ with nothing to prove. For the reverse inclusion we must show that every real x is an adherent point of $\mathbb{Q}$ — that is, given any $\varepsilon > 0$, exhibit a rational within ε of x.

So let $x \in \mathbb{R}$ and $\varepsilon > 0$.

Step 1: choose a fine enough mesh. By (F2) there is a positive integer N with $1/\varepsilon \le N$. As N and ε are positive, this rearranges to

$$\frac{1}{N} \;\le\; \varepsilon.$$

Step 2: snap x to the grid of multiples of $1/N$. By (F10) put $k := \lfloor Nx \rfloor$, the unique integer with

$$k \;\le\; Nx \;<\; k + 1.$$

Dividing through by $N > 0$ preserves both inequalities:

$$\frac{k}{N} \;\le\; x \;<\; \frac{k+1}{N}.$$

Step 3: the approximation is good enough. The left inequality gives $x - k/N \ge 0$, so $|x - k/N| = x - k/N$; and the right inequality gives

$$x - \frac{k}{N} \;<\; \frac{k+1}{N} - \frac{k}{N} \;=\; \frac{1}{N}.$$

Combining with Step 1,

$$\left|x - \frac{k}{N}\right| \;<\; \frac{1}{N} \;\leq\; \varepsilon,$$

and k/N is rational, being a ratio of the integers k and $N \neq 0$.

As $\varepsilon > 0$ was arbitrary, x is an adherent point of $\mathbb{Q}$. Hence $\mathbb{R} \subseteq \overline{\mathbb{Q}}$, and $\overline{\mathbb{Q}} = \mathbb{R}$.

$\overline{\mathbb{R}} = \mathbb{R}$. By (D5) every adherent point is a real number, so $\overline{\mathbb{R}} \subseteq \mathbb{R}$; and $\mathbb{R} \subseteq \overline{\mathbb{R}}$ by the free inclusion.

$\overline{\emptyset} = \emptyset$. Let $x \in \mathbb{R}$ be arbitrary and take $\varepsilon := 1$. Being 1-adherent to $\emptyset$ would require some $y \in \emptyset$ with $|x - y| \leq 1$, and no y belongs to $\emptyset$ at all. So x is not 1-adherent to $\emptyset$, hence not an adherent point of it. As x was arbitrary, $\overline{\emptyset}$ has no elements.

Remark. The first two parts and the third are opposite extremes, and the proofs show why. For $\mathbb{Z}$ and $\mathbb{N}$ the points of the set are spread out, leaving a gap of positive width around any x outside it — and one such gap is all a witness $(*)$ needs. For $\mathbb{Q}$ no such gap exists anywhere: the grid of multiples of $1/N$ can be made as fine as we please, so the approximation $x \approx \lfloor Nx \rfloor / N$ succeeds for every ε. That approximation is the standard one, and reappears whenever a real number must be approximated by rationals.

Question (7). Let E be a non-empty bounded subset of $\mathbb{R}$, and let $S := \sup(E)$ be the least upper bound of E. (Note that S is a real number, since a non-empty set which is bounded above has a real least upper bound.) Show that S is an adherent point of E, and is also an adherent point of $\mathbb{R} \setminus E$.

S is adherent to E. Let $\varepsilon > 0$. Since E is non-empty and bounded above, (F6) supplies $y \in E$ with $y > S - \varepsilon$; and $y \leq S$ because S is an upper bound for E, again by (F6). Hence

$$0 \;\leq\; S - y \;<\; \varepsilon,$$

so $|S - y| \leq \varepsilon$ with $y \in E$. As $\varepsilon > 0$ was arbitrary, S is an adherent point of E by (D5).

S is adherent to $\mathbb{R} \setminus E$. Let $\varepsilon > 0$ and put $z := S + \varepsilon/2$. Since $z > S$ and S is an upper bound for E, the point z cannot lie in E; so $z \in \mathbb{R} \setminus E$. And

$$|S - z| \;=\; \frac{\varepsilon}{2} \;\leq\; \varepsilon.$$

So S is ε-adherent to $\mathbb{R} \setminus E$ for every $\varepsilon > 0$.

Remark. The second half holds whatever E is: the points just above the supremum are never in E. Together the two halves say that S lies in the closure of E and in the closure of its complement — that is, S is a boundary point of E.

Part B. Further questions

Question (8). Let $A, B \subseteq \mathbb{R}$ be closed.

(a) Show that $A \cup B$ is closed.

(b) Deduce that $X_1 \cup \cdots \cup X_n$ is closed whenever $X_1, \ldots, X_n$ are closed and $n \geq 1$.

(c) Give closed sets $X_1, X_2, X_3, \ldots$ whose union is not closed. Why does this not contradict the fact that the intersection of *any* family of closed sets is closed?

(a) By (D6) we must show $\overline{A \cup B} = A \cup B$.

The inclusion $A \cup B \subseteq \overline{A \cup B}$. This holds for any set: each $x \in A \cup B$ is ε-adherent to $A \cup B$ for every $\varepsilon > 0$, taking $y := x$.

The inclusion $\overline{A \cup B} \subseteq A \cup B$. Let x be an adherent point of $A \cup B$, and suppose, for contradiction, that $x \notin A \cup B$. Then $x \notin A$, and since A is closed, x is not an adherent point of A. Negating (D5), there is $\varepsilon_1 > 0$ such that x is not ε_1-adherent to A, that is,

$$|x - y| \;>\; \varepsilon_1 \qquad \text{for every } y \in A.$$

In the same way, since $x \notin B$ and B is closed, there is $\varepsilon_2 > 0$ with $|x - y| > \varepsilon_2$ for every $y \in B$.

Put $\varepsilon := \min(\varepsilon_1, \varepsilon_2)$, which is positive. Since x is an adherent point of $A \cup B$, it is ε-adherent to $A \cup B$: there is $y \in A \cup B$ with $|x - y| \leq \varepsilon$. If $y \in A$ then $|x-y| \leq \varepsilon \leq \varepsilon_1$, contradicting the choice of ε_1; and if $y \in B$ then $|x-y| \leq \varepsilon \leq \varepsilon_2$, contradicting the choice of ε_2. Either way we have a contradiction, so $x \in A \cup B$.

Hence $\overline{A \cup B} = A \cup B$, and $A \cup B$ is closed.

(b) By induction on n. For $n = 1$ the union is X_1, which is closed. Suppose that every union of n closed sets is closed, and let $X_1, \ldots, X_{n+1}$ be closed. Then

$$X_1 \cup \cdots \cup X_{n+1} \;=\; \big(X_1 \cup \cdots \cup X_n\big) \cup X_{n+1}$$

is the union of two sets which are closed, the first by the induction hypothesis and the second by assumption. By (a) it is closed.

(c) *The example.* Let $X_n := \{1/n\}$ for $n \geq 1$.

Each X_n is closed. More generally, every singleton $\{c\}$ is closed. Let x be an adherent point of $\{c\}$ and let $\varepsilon > 0$. By (D5) there is $y \in \{c\}$ with $|x - y| \leq \varepsilon$, and the only such y is c itself; so

$$|x - c| \;\leq\; \varepsilon \qquad \text{for every } \varepsilon > 0.$$

Suppose $x \neq c$. Then $|x - c| > 0$ by (F1), so $\varepsilon := |x - c|/2$ is a legitimate choice, and it gives $|x - c| \leq |x - c|/2$, hence $|x - c| \leq 0$ and therefore $|x - c| = 0$. By (F1) this forces $x = c$, contradicting $x \neq c$. Hence $x = c$, so $\overline{\{c\}} = \{c\}$ and $\{c\}$ is closed by (D6).

The union is not closed. The union of the X_n is

$$U \;:=\; \bigcup_{n \geq 1} X_n \;=\; \Big\{ 1, \tfrac{1}{2}, \tfrac{1}{3}, \tfrac{1}{4}, \ldots \Big\}.$$

The point 0 is an adherent point of U. Indeed, let $\varepsilon > 0$. By (F2) there is a positive integer N with $1/\varepsilon \leq N$, which rearranges to $1/N \leq \varepsilon$; and $1/N \in U$ with

$$|0 - 1/N| \;=\; 1/N \;\leq\; \varepsilon.$$

So 0 is ε-adherent to U for every $\varepsilon > 0$, that is, $0 \in \overline{U}$ by (D5). But $0 \notin U$, since every element of U is of the form $1/n$ with $n \geq 1$ and hence is strictly positive. Therefore $U \neq \overline{U}$, and U is not closed by (D6).

Why this is not a contradiction. The result quoted concerns *intersections*, whereas the example above is a *union*. The two operations behave quite differently, and nothing in the example says anything at all about an intersection.

One can check directly that for this very family the quoted result does hold. A real number lying in every X_n would have to equal 1 (to lie in X_1) and also $\frac{1}{2}$ (to lie in X_2), which is impossible; hence

$$\bigcap_{n \geq 1} X_n \;=\; \emptyset.$$

And $\emptyset$ is indeed closed: for any $x \in \mathbb{R}$, the condition "there is $y \in \emptyset$ with $|x - y| \leq 1$" requires some y to belong to $\emptyset$, and none does, so no real number is even 1-adherent to $\emptyset$. Thus $\overline{\emptyset} = \emptyset$, exactly as predicted.

Remark. Closedness survives intersections of arbitrarily many sets, but unions of only finitely many. A union of two closed sets is closed, and by induction so is any finite union; this example shows the induction cannot be pushed to infinitely many sets.

Question (9). The alternating series test, stated as (F12), requires the sequence (a_n) to be decreasing as well as non-negative. Show that this hypothesis cannot be dropped: exhibit a sequence $(a_n)_{n=1}^{\infty}$ with $a_n \geq 0$ and $a_n \to 0$ for which $\sum_{n=1}^{\infty}(-1)^n a_n$ diverges. Show also that $\sum_{n=1}^{\infty}(-1)^n/n$ does converge.

The convergent example. Put $b_n := 1/n$ for $n \geq 1$. Then $b_n \geq 0$; and $b_n \geq b_{n+1}$ since $0 < n \leq n+1$; and $b_n \to 0$, since given $\varepsilon > 0$, (F2) supplies a positive integer N with $1/\varepsilon \leq N$, and then $|b_n - 0| = 1/n \leq 1/N \leq \varepsilon$ for all $n \geq N$. So (b_n) is non-negative, decreasing and null, and (F12) gives the convergence of $\sum_{n=1}^{\infty}(-1)^n/n$.

The counterexample. Define, for $n \geq 1$,

$$a_n := \begin{cases} 1/n, & n \text{ even}, \\ 2^{-n}, & n \text{ odd}. \end{cases}$$

$a_n \geq 0$ *and* $a_n \to 0$. Non-negativity is clear. We claim $a_n \leq 1/n$ for every n: for even n this is an equality, and for odd n it says $2^{-n} \leq 1/n$, that is, $n \leq 2^n$, which holds by induction ($1 \leq 2$, and $n+1 \leq 2^n + 1 \leq 2^n + 2^n = 2^{n+1}$). Now given $\varepsilon > 0$, (F2) gives a positive integer N with $1/\varepsilon \leq N$, and for $n \geq N$ we get $|a_n - 0| = a_n \leq 1/n \leq 1/N \leq \varepsilon$.

(a_n) *is not decreasing.* $a_3 = 1/8$ while $a_4 = 1/4 > a_3$.

The series diverges. Let $S_N := \sum_{n=1}^{N}(-1)^n a_n$ and split the sum by parity: the even n contribute $+1/n$ and the odd n contribute -2^{-n}, so

$$S_N \;=\; \sum_{\substack{n \leq N \\ n \text{ even}}} \frac{1}{n} \;-\; \sum_{\substack{n \leq N \\ n \text{ odd}}} \frac{1}{2^n}.$$

The subtracted sum is at most $\sum_{n=1}^{N} 2^{-n} = 1 - 2^{-N} < 1$, the identity following by induction on N. The even n with $1 \leq n \leq N$ are exactly the numbers $n = 2k$ with k an integer, $1 \leq k \leq \lfloor N/2 \rfloor$: indeed $2k \leq N$ says $k \leq N/2$, and for an integer k this is equivalent to $k \leq \lfloor N/2 \rfloor$, since $\lfloor N/2 \rfloor \leq N/2$ by (F10) while $k > \lfloor N/2 \rfloor$ would force $k \geq \lfloor N/2 \rfloor + 1 > N/2$. Hence

$$S_N \;\geq\; \sum_{k=1}^{\lfloor N/2 \rfloor} \frac{1}{2k} \;-\; 1 \;=\; \frac{1}{2} \sum_{k=1}^{\lfloor N/2 \rfloor} \frac{1}{k} \;-\; 1.$$

By (F11) the partial sums of $\sum_k 1/k$ are not bounded above, and $\lfloor N/2 \rfloor \to \infty$, so the right-hand side exceeds any prescribed bound for suitable N. Hence (S_N) is not bounded above, so it is not a bounded sequence, and by (F5) it does not converge. By (D2) the series diverges.

Remark. The example is built so that the positive terms are large enough to diverge on their own while the negative terms are summable. The decreasing hypothesis in (F12) is exactly what rules out this imbalance.

Question (10). Let $(c_n)_{n=1}^{\infty}$ be the sequence of positive numbers defined by

$$c_n := 2^{-2\lfloor n/2 \rfloor} \qquad (n \geq 1).$$

Prove that

$$\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \;=\; \frac{1}{4}, \qquad \lim_{n\to\infty} c_n^{1/n} \;=\; \frac{1}{2}, \qquad \limsup_{n\to\infty} \frac{c_{n+1}}{c_n} \;=\; 1.$$

The first terms are $c_1 = 1$, $c_2 = c_3 = \frac{1}{4}$, $c_4 = c_5 = \frac{1}{16}$, and so on; every c_n is positive.

The floor values. By (F10), $\lfloor n/2 \rfloor$ is the unique integer with $\lfloor n/2 \rfloor \leq n/2 < \lfloor n/2 \rfloor + 1$, so c_n is well defined. We compute it, and the uniqueness in (F10) is what lets us do so: to identify $\lfloor t \rfloor$ it is enough to exhibit one integer k with $k \leq t < k+1$.

If $n = 2k$ is even, with $k \geq 0$ an integer, then $n/2 = k$, and k is an integer satisfying $k \leq k < k+1$; hence $\lfloor n/2 \rfloor = k$. Also $(n+1)/2 = k + \frac{1}{2}$, and the integer k satisfies $k \leq k + \frac{1}{2} < k+1$; hence $\lfloor (n+1)/2 \rfloor = k$ as well. So

$$\lfloor (n+1)/2 \rfloor - \lfloor n/2 \rfloor \;=\; 0 \qquad (n \text{ even}).$$

If $n = 2k+1$ is odd, then $n/2 = k + \frac{1}{2}$ and $k \leq k + \frac{1}{2} < k+1$, so $\lfloor n/2 \rfloor = k$; while $(n+1)/2 = k+1$ is itself an integer, and $k+1 \leq k+1 < k+2$, so $\lfloor (n+1)/2 \rfloor = k+1$. So

$$\lfloor (n+1)/2 \rfloor - \lfloor n/2 \rfloor \;=\; 1 \qquad (n \text{ odd}).$$

In particular $c_{2k} = c_{2k+1} = 2^{-2k}$, which is the pattern listed above.

The ratios. Write $r_n := c_{n+1}/c_n$ for $n \geq 1$. Since $c_n = 2^{-2\lfloor n/2 \rfloor}$,

$$r_n \;=\; \frac{c_{n+1}}{c_n} \;=\; \frac{2^{-2\lfloor (n+1)/2 \rfloor}}{2^{-2\lfloor n/2 \rfloor}} \;=\; 2^{\,-2(\lfloor (n+1)/2 \rfloor \,-\, \lfloor n/2 \rfloor)}.$$

By the two displays above the exponent difference is 0 when n is even and 1 when n is odd, so

$$r_n \;=\; 2^0 \;=\; 1 \quad (n \text{ even}), \qquad r_n \;=\; 2^{-2} \;=\; \tfrac{1}{4} \quad (n \text{ odd}).$$

So the ratio sequence (r_n) takes only the values 1 and $\frac{1}{4}$, each at infinitely many indices. Hence $\{r_n : n \geq N\} = \{\frac{1}{4}, 1\}$ for every N, so $r_N^+ = 1$ and $r_N^- = \frac{1}{4}$, and by (D4) $\limsup_{n\to\infty} r_n = \inf\{1\} = 1$ while $\liminf_{n\to\infty} r_n = \sup\{\frac{1}{4}\} = \frac{1}{4}$.

The roots. If $n = 2k$ is even then $c_n = 2^{-2k} = 2^{-n}$, and $\frac{1}{2}$ is a non-negative real whose nth power is 2^{-n}, so $c_n^{1/n} = \frac{1}{2}$ by (F9).

If $n = 2k+1$ is odd then $c_n = 2^{-2k} = 2^{-(n-1)}$. Put $v := \frac{1}{2} \cdot 2^{1/n} > 0$. Then, using (F9),

$$v^n \;=\; 2^{-n}\big(2^{1/n}\big)^n \;=\; 2^{-n} \cdot 2 \;=\; 2^{-(n-1)} \;=\; c_n,$$

so $c_n^{1/n} = v = \frac{1}{2} \cdot 2^{1/n}$, again by the uniqueness in (F9).

The limit. Let $\varepsilon > 0$. By (F8) with $c := 2$ there is N with $|2^{1/n} - 1| \leq 2\varepsilon$ for all $n \geq N$. For $n \geq N$: if n is even then $|c_n^{1/n} - \frac{1}{2}| = 0 \leq \varepsilon$; and if n is odd then by (F1),

$$\left|c_n^{1/n} - \tfrac{1}{2}\right| \;=\; \tfrac{1}{2}\left|2^{1/n} - 1\right| \;\leq\; \varepsilon.$$

So $\lim_{n\to\infty} c_n^{1/n} = \frac{1}{2}$ by (D1).

Collecting the three. We have shown

$$\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \;=\; \frac{1}{4}, \qquad \lim_{n\to\infty} c_n^{1/n} \;=\; \frac{1}{2}, \qquad \limsup_{n\to\infty} \frac{c_{n+1}}{c_n} \;=\; 1,$$

which is what was asked.

Remark. For every sequence of positive numbers one has

$$\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \;\leq\; \liminf_{n\to\infty} c_n^{1/n} \;\leq\; \limsup_{n\to\infty} c_n^{1/n} \;\leq\; \limsup_{n\to\infty} \frac{c_{n+1}}{c_n},$$

and the three values just computed show that both outer inequalities can be strict: here they read $\frac{1}{4} < \frac{1}{2}$ and $\frac{1}{2} < 1$. This is the precise sense in which the root test is stronger than the ratio test. For this sequence the root quantity $\frac{1}{2}$ is less than 1, which settles the convergence of $\sum c_n$; the ratio quantity $\limsup c_{n+1}/c_n$ equals 1, which settles nothing.